

Lipschitz condition

Def: Let $D \subseteq \mathbb{R}^2$ be an open rectangle and $f: D \rightarrow \mathbb{R}$. The function f is said to be bounded on D if $\exists$ a positive no. M s.t. $|f(x, y)| \leq M$ for all $(x, y) \in D$.

Result: Let $\bar{D} = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$. Any continuous function on $\bar{D}$ is bounded.

Def: A function f is said to satisfy Lipschitz condition w.r to y in D if $\exists$ a constant M s.t.

$$|f(x, y_1) - f(x, y_2)| \leq M |y_1 - y_2|$$

for all $(x, y_1), (x, y_2) \in D$.

The constant M is called Lipschitz constant.

Remark: If f satisfies Lipschitz condition w.r to y in D then for each x the resultant function of y is continuous function of y for all $(x, y) \in D$.

($\phi_x(y) = f(x, y)$ is continuous in $y \neq x$, $(x, y) \in D$)

May not be continuous in x .

Ex: $f(x, y) = y + [x]$, $[]$ - greatest integer function

$$\begin{aligned} f(x, y_1) - f(x, y_2) &= y_1 + [x] - y_2 - [x] \\ &= y_1 - y_2 \end{aligned}$$

$$|f(x, y_1) - f(x, y_2)| = |y_1 - y_2|.$$

$\Rightarrow f$ is Lipschitz in y , hence continuous in y , but it is not continuous in x .

Remark: The condition of Lipschitz continuity in y implies nothing about continuity of f . Moreover, continuity of f does not imply Lipschitz condition in y .

Ex: $f(x, y) = \sqrt{|y|}$.

Clearly, f is continuous on $\mathbb{R}^2$.

$$f(x, y_1) - f(x, y_2) = \sqrt{|y_1|} - \sqrt{|y_2|}$$

Take $y_2 = 0$, $y_1 \neq 0$.

$$f(x, y_1) - f(x, 0) = \sqrt{|y_1|}$$

$$\begin{aligned} \therefore \frac{|f(x, y_1) - f(x, 0)|}{|y_1 - 0|} &= \frac{\sqrt{|y_1|}}{|y_1|} \\ &= \frac{1}{\sqrt{|y_1|}} \rightarrow \infty \text{ as } y_1 \rightarrow 0. \end{aligned}$$

Thus, f is not Lipschitz on any domain containing $(0, 0)$.

Result: If f is s.t. $\frac{\partial f}{\partial y}$ exist and bounded for all $(x, y) \in D$,

then f is Lipschitz in y .

Outline: By mean value theorem,

$$f(x, y_1) - f(x, y_2) = \frac{\partial f}{\partial y}(x, c)(y_1 - y_2), \quad y_1 < c < y_2.$$

$$\begin{aligned} |f(x, y_1) - f(x, y_2)| &= \left| \frac{\partial f}{\partial y}(x, c) \right| |y_1 - y_2| \\ &\leq M |y_1 - y_2| \quad \left(\because \left| \frac{\partial f}{\partial y} \right| \leq M \right) \end{aligned}$$

Ex: $f(x, y) = y^2$, $D: |x| < a, |y| < b$.

$$\frac{\partial f}{\partial y} = 2y, \quad \left| \frac{\partial f}{\partial y} \right| = |2y| \leq 2b.$$

$\Rightarrow f$ is Lipschitz in y .

Ex $f(x, y) = x|y|$ $D: |x| < a, |y| < b$.

Here, $\frac{\partial f}{\partial y}$ does not exist at $y=0$. But

$$\begin{aligned} f(x, y_1) - f(x, y_2) &= x|y_1| - x|y_2| \\ |f(x, y_1) - f(x, y_2)| &= |x|y_1| - x|y_2|| \\ &= |x| ||y_1| - |y_2|| \\ &\leq a ||y_1| - |y_2|| \\ &\leq a |y_1 - y_2| \quad (?) \end{aligned}$$

$\Rightarrow f$ is Lipschitz in y .

Thus, the above result gives only sufficient condition.

Ex. 1 $f(x, y) = \sin y$, $D: |x| \leq a, |y| \leq b$.
Lipschitz in y .

Ex. 2 $f(x, y) = \frac{1}{y^2}$ on $D: |x| \leq a, y \in [b, \infty), b > 0$.

Is f Lipschitz in y ?

Ex. 3 $f(x, y) = y^{1/3}$, $D: |x| \leq a, |y| \leq b$.
Is f Lipschitz in y ?

Existence - Uniqueness theorem (Picard):

Let $D: |x - x_0| < a, |y - y_0| < b$.

IVP: $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$. — ①

Existence: (1) If f is continuous for all $(x, y) \in D$ and
(2) f is bounded on D , i.e., $|f(x, y)| \leq K, \forall (x, y) \in D$,
K - constant.

Then IVP (1) has at least one solution y defined for all
 x in the interval $|x - x_0| < \alpha$, where
 $\alpha = \min \left\{ a, \frac{b}{K} \right\}$.

Uniqueness: In addition to the above condition, if f satisfies
the Lipschitz condition w.r to y in D then the
solution y defined at least for all x in the interval
 $|x - x_0| < \alpha$ with α defined above is unique.

Ex. 1 $\frac{dy}{dx} = x|y|$, $y(0) = 0$, $|x| < a, |y| < b$.

Existence: (1) $f(x, y) = x|y|$ is continuous
(2) $|f(x, y)| = |x||y| \leq |x||b| < ab$.

$\therefore$ The solution exists for all x in $|x| < \alpha = \min \left\{ a, \frac{b}{ab} \right\}$
 $= \min \left\{ a, \frac{1}{a} \right\}$

Uniqueness: By the above, f is Lipschitz in y in D .

Thus, the solution is unique.

Ex.2 $\frac{dy}{dx} = 2x, \quad y(0) = 1, \quad |x| < a, \quad |y-1| < b.$

The solution exists and unique.

Ex.3 $\frac{dy}{dx} = 1+y^2, \quad y(0) = 0, \quad |x| < 5, \quad |y| < 3.$

Solⁿ $f(x, y) = 1+y^2$ is continuous on $|x| < 5, |y| < 3.$

$$|f(x, y)| = |1+y^2| \leq 1+|y|^2 \leq 1+9 = 10.$$

$\therefore$ The solution exists.

$$\frac{\partial f}{\partial y} = 2y \quad \& \quad \left| \frac{\partial f}{\partial y} \right| \leq 2|y| \leq 6.$$

$\Rightarrow$ Thus, f is Lipschitz in y .

Hence, solution is unique.

Interval of the solution: $|x| < \alpha = \min \left\{ 5, \frac{3}{10} \right\} = \frac{3}{10}.$

Ex.4: $\frac{dy}{dx} = y^2, \quad y(1) = -1.$ Verify the existence and

uniqueness theorem. If solution exists, find α in the existence & uniqueness theorem.

Sol: $f(x, y) = y^2$ is continuous and bounded on any domain $D: |x-1| \leq a + |y+1| \leq b.$

$$|f(x, y)| = |y^2| = |y|^2 \leq |(-b-1)|^2 \leq (b+1)^2$$

$(\because -b \leq y+1 \leq b \Rightarrow -1-b \leq y \leq -1+b \leq 1+b)$

Since $\frac{\partial f}{\partial x} = 0$ and $\left| \frac{\partial f}{\partial y} \right| = 2|y| \leq 2|1+b|,$

thus f is Lipschitz in y on D .

Hence, solution exists and unique.

Interval of the solution $|x-1| < \alpha = \min \left\{ a, \frac{b}{(b+1)^2} \right\}.$

Note: we can further simplify the interval of the solution.

Let $g(b) = \frac{b}{(b+1)^2}.$

Check: The maximum value of $g(b)$ for $b > 0$ occurs at $b=1$.

We have $g(1) = \frac{1}{4}$.

If $a \geq \frac{1}{4}$, $g(b) = \frac{b}{(b+1)^2} \leq \frac{1}{4} \leq a$ for all $b > 0$ and

$\alpha = \min \{a, g(b)\} = g(b) \leq \frac{1}{4}$. For $b=1$, $\alpha = \frac{1}{4}$.

If $a < \frac{1}{4}$, then $\alpha = \min \{a, g(b)\} < \frac{1}{4}$.

Thus, the best possible α from the theorem gives that the IVP has a unique solution in $|x-1| \leq \frac{1}{4}$, i.e., $\frac{3}{4} \leq x \leq \frac{5}{4}$.

Ex: Verify the existence and uniqueness theorem for

(i) $\frac{dy}{dx} = y^{1/3}$, $y(0)=0$, $|x| < a$, $|y| < b$.

(ii) $\frac{dy}{dx} = \sqrt{|y|}$, $y(0)=0$, $|x| < a$, $|y| < b$.

(iii) $y' = 2\sqrt{y}$, $y(0)=0$, $|x-1| < a$, $|y| < b$.

(iv) $y' = x - y^2$, $y(0)=1$, $|x| < a$, $|y-1| < b$.

Remark: (i) The theorem guarantees existence and uniqueness only in a very small interval.

(ii) The theorem does not give the largest interval where the solution is unique.

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